



Satyaansh Softech



Project Based Learning



Something new for everyone — beginner or pro



100% Project-Based Learning



Every video = Contains assignments



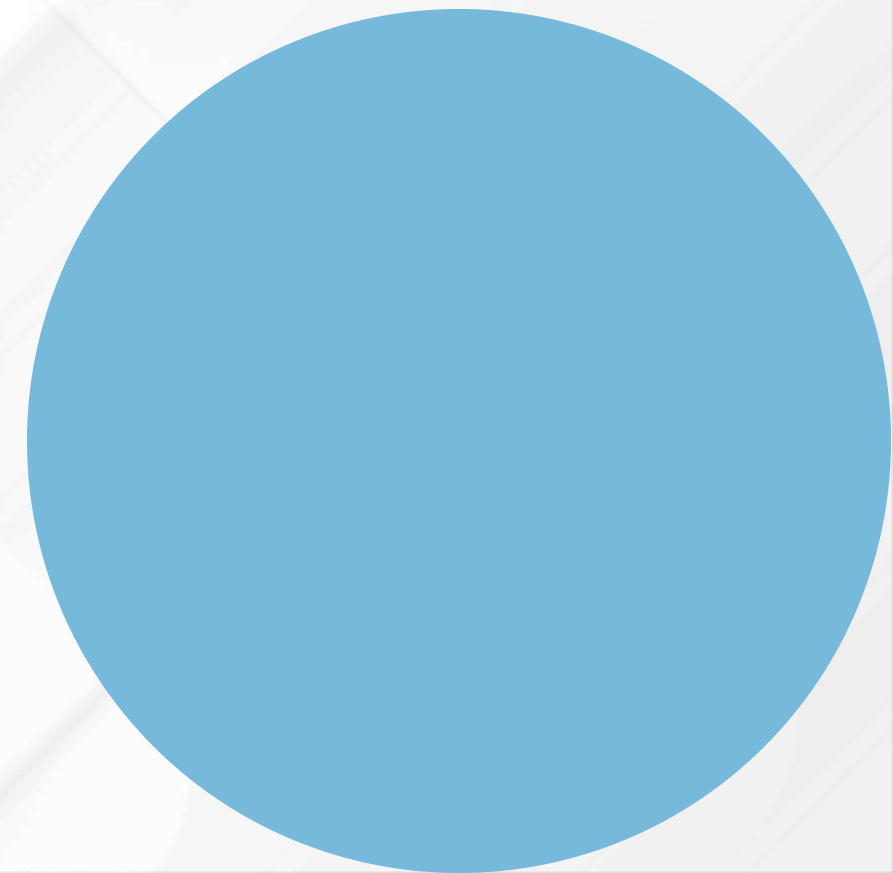
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Shallow & Deep Copy in Js

By Satyawan Panchal



Outline

01 Shallow Copy

02 Deep Copy

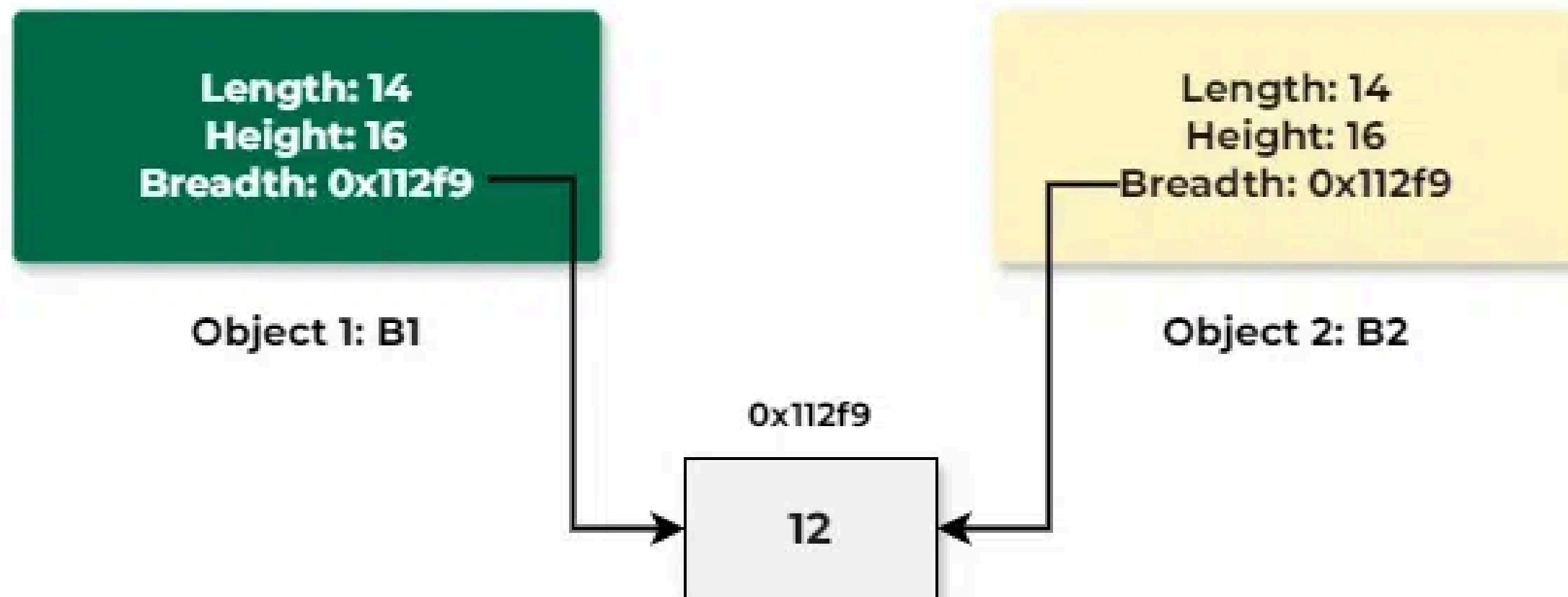
03 Better Approach

04 How to make deep copy

Along with examples

Shallow Copy

Change in copy → Original Changed

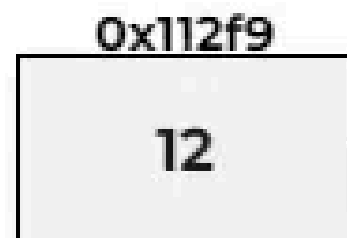


Deep Copy

Changed in copied → No Change in Original



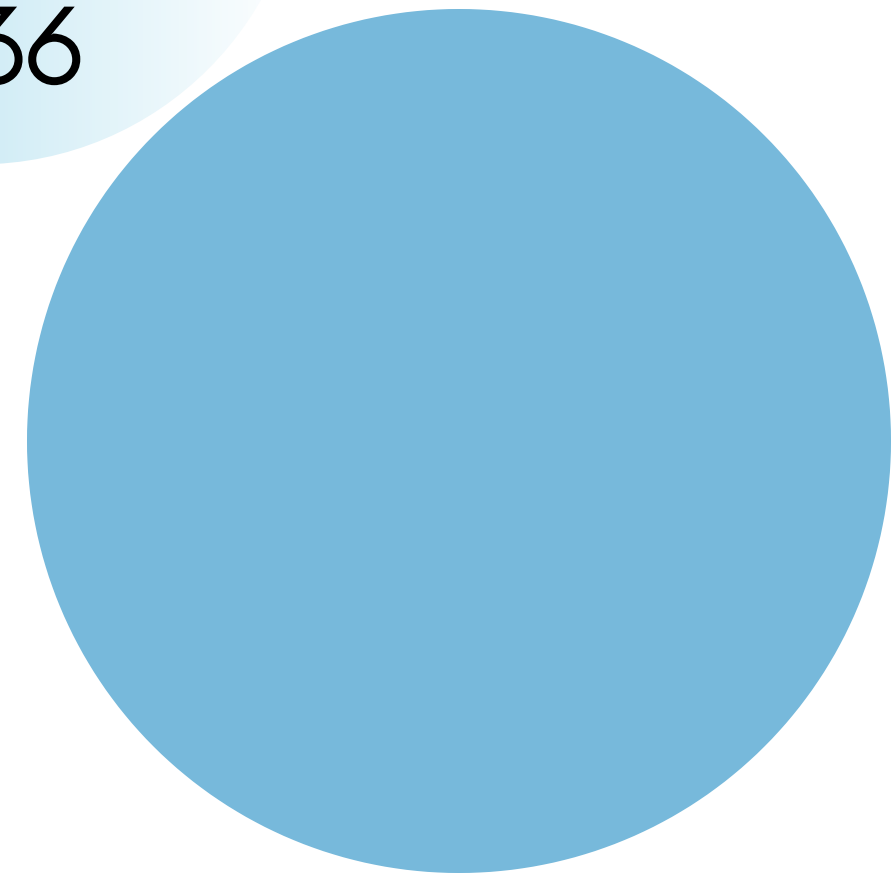
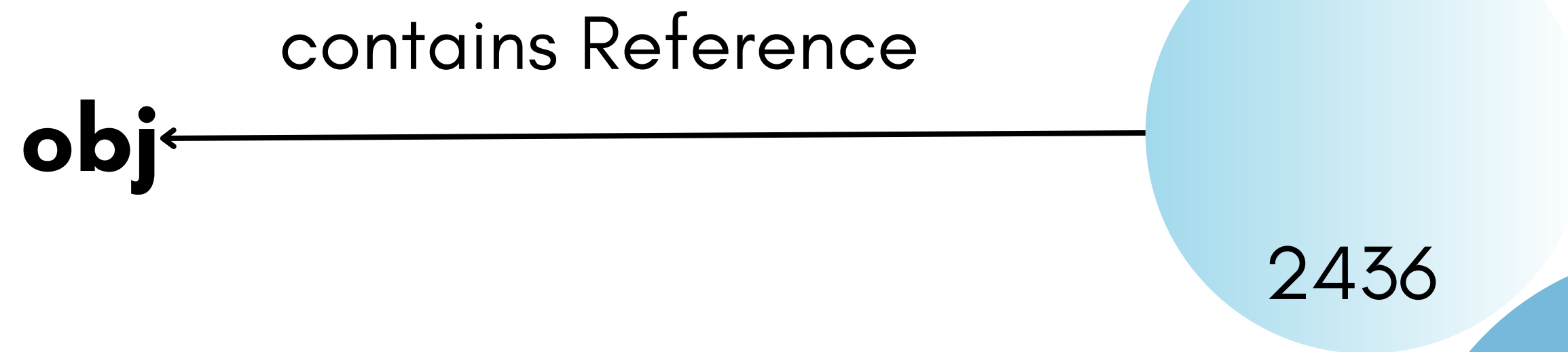
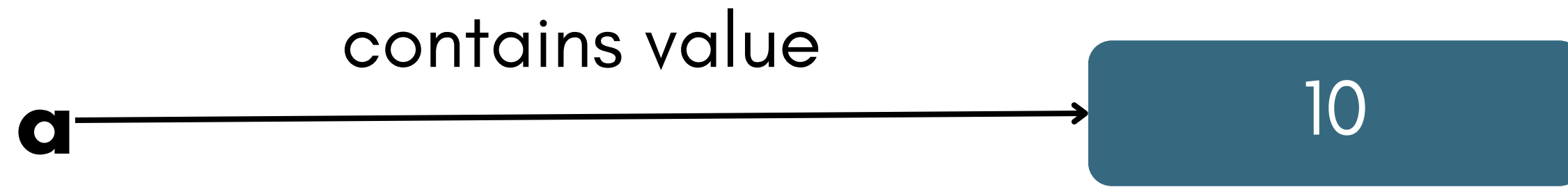
Object 1: B1



Object 2: B2



a=10



copyObj={...orgObj}

copyObj

```
name: "Rinku",  
age: 25,  
address: {  
  city: "Panipat",  
  pincode: "132101"  
}
```

orgObj

```
name: "Rinku",  
age: 25,  
address: {  
  city: "Panipat",  
  pincode: "132101"  
}
```

`copyObj.address.city= "Panipat"`

`console.log(orgObj.address.city)`
Panipat

{
 city: "Panipat",
 pincode: "132101"
}

2436



Which one is useful and Why?

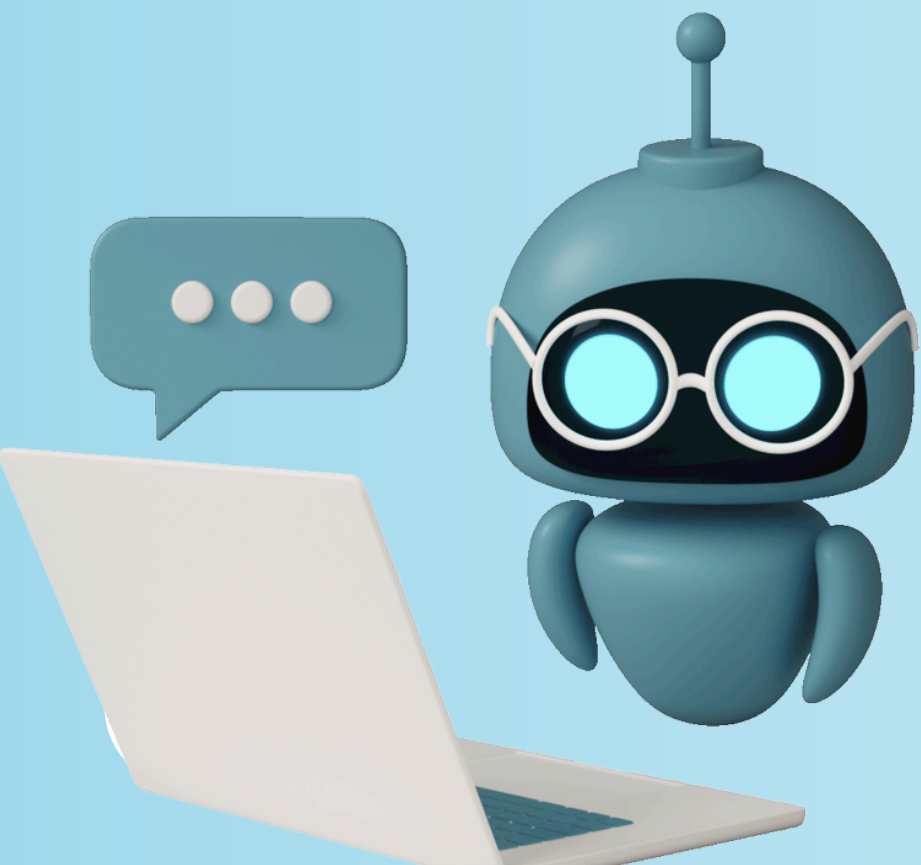


Shallow Copy



Useful When:

- Copy a simple object/array
- (No nested structures).



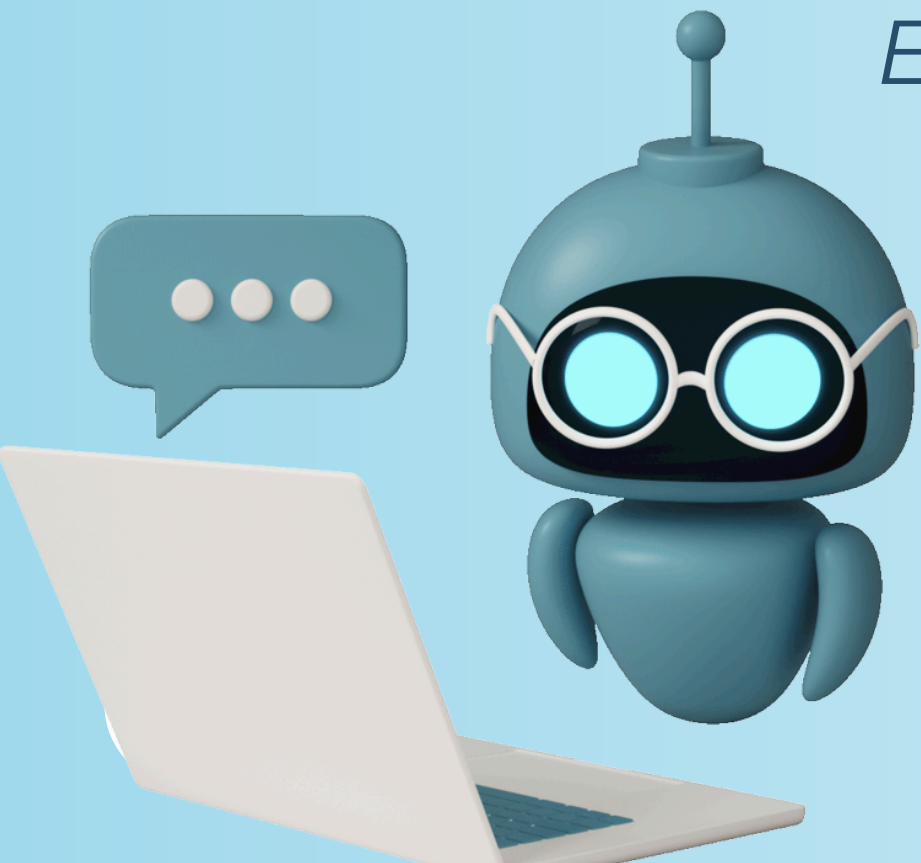
Deep Copy



Useful When:

- Need to safely modify data
- Without risk of changing the original..

*Examples - Handling form data, cart items,
or database responses.*

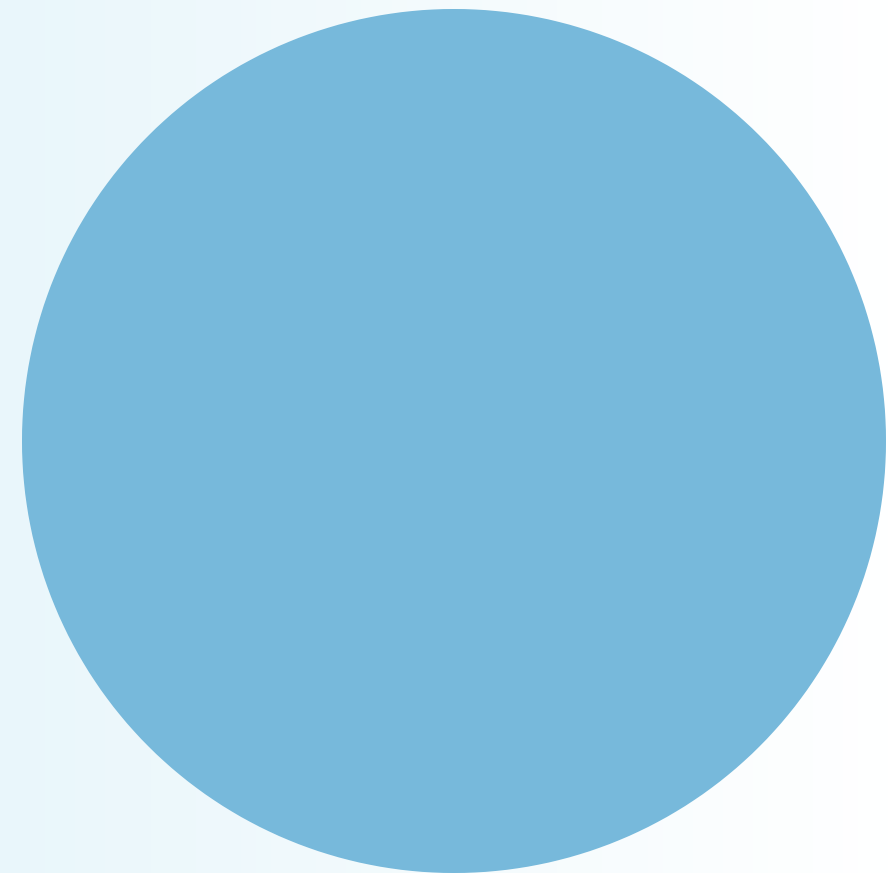
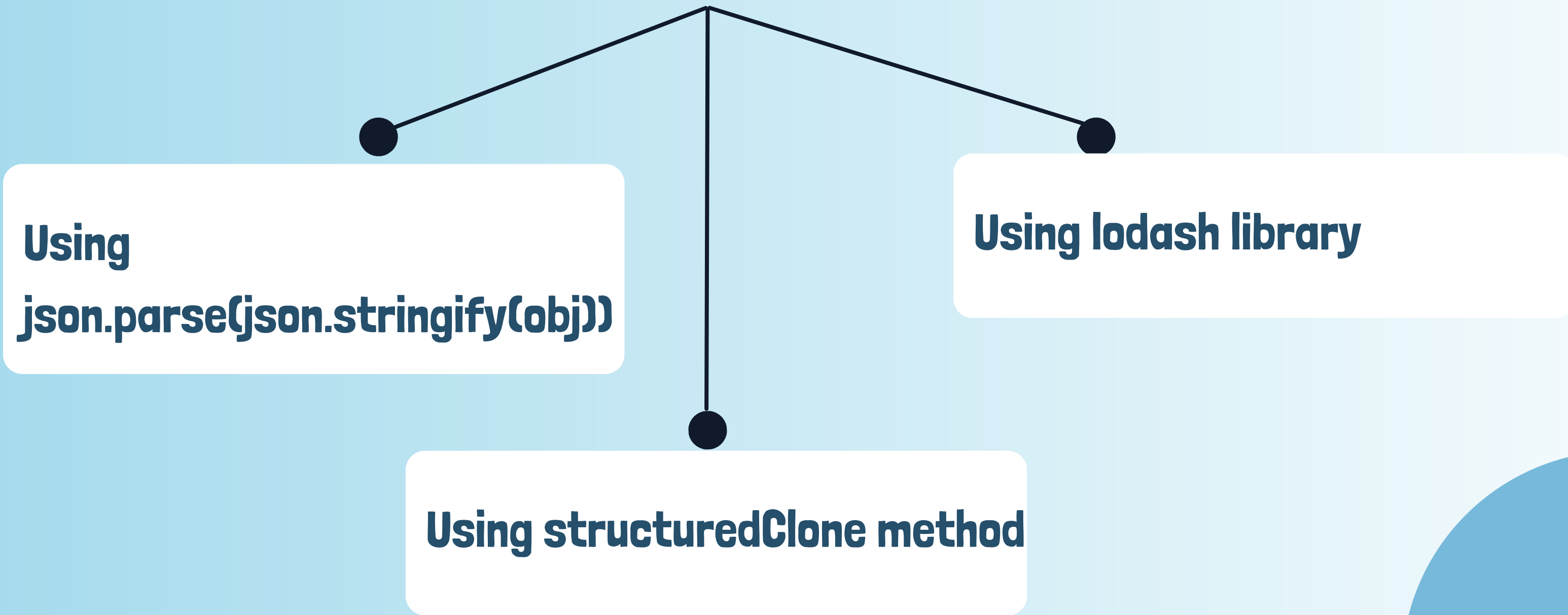


Problem with shallow copy



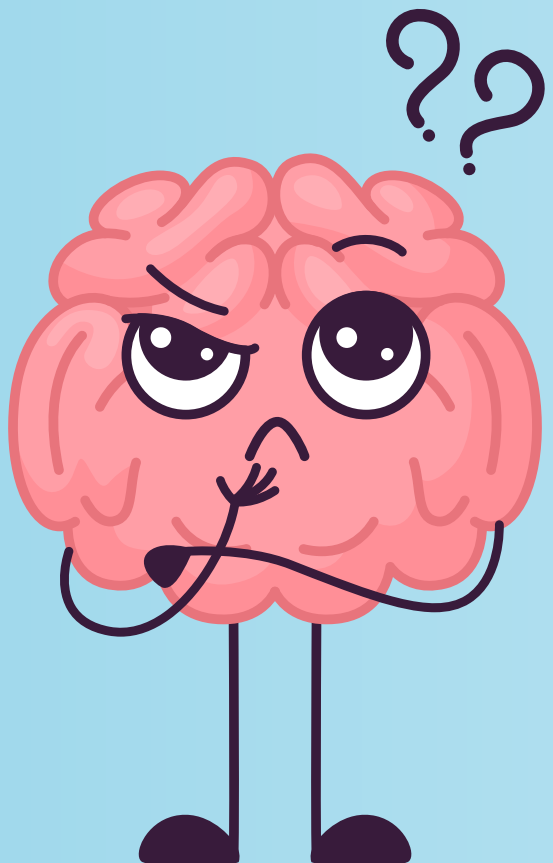
✗ Linked references may be updated unintentionally.

HOW TO MAKE A DEEP COPY



PROBLEM WITH JSON.parse()

- Loses non-JSON data types
- Functions, undefined, Symbol, Date, Map, Set, BigInt, RegExp are not preserved.



PROBLEM WITH PARSING(EX.)



```
const obj = {  
  date: new Date(),  
  regex: /abc/g,  
  fn: () => 42  
};
```

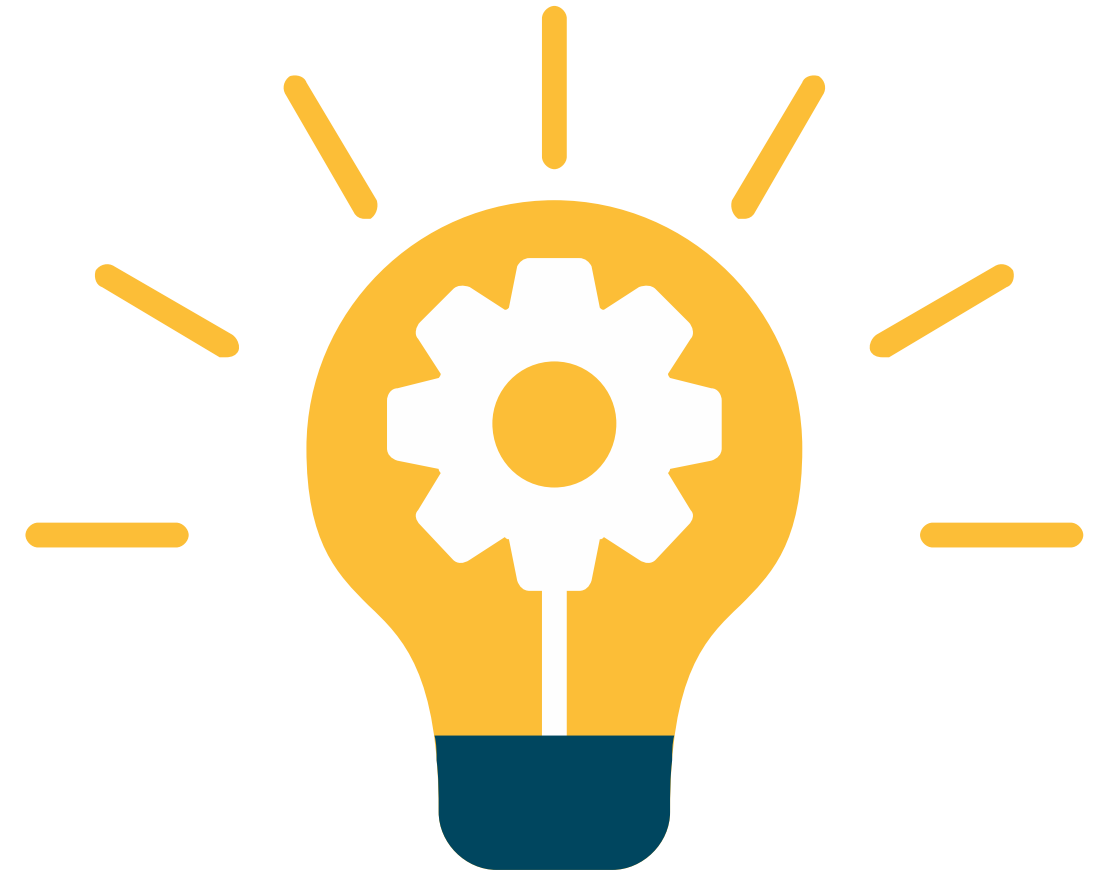
```
const copy = JSON.parse(JSON.stringify(obj));
```

```
console.log(copy.date); // string, not Date
```

```
console.log(copy.regex); // {} (lost)
```

```
console.log(copy.fn); // undefined
```

SOLUTIONS



SOLUTION 2

→ Use **structuredClone(obj)** (Modern JS)

PROS:

Handles more data types correctly:

- Date, Map, Set, RegExp, ArrayBuffer, Blob, File, ImageBitmap, etc.

HOW TO USE ?

```
const copy = structuredClone(obj);
```

PROBLEM

Not supported by in old browsers.

SOLUTION 3

Use lodash library

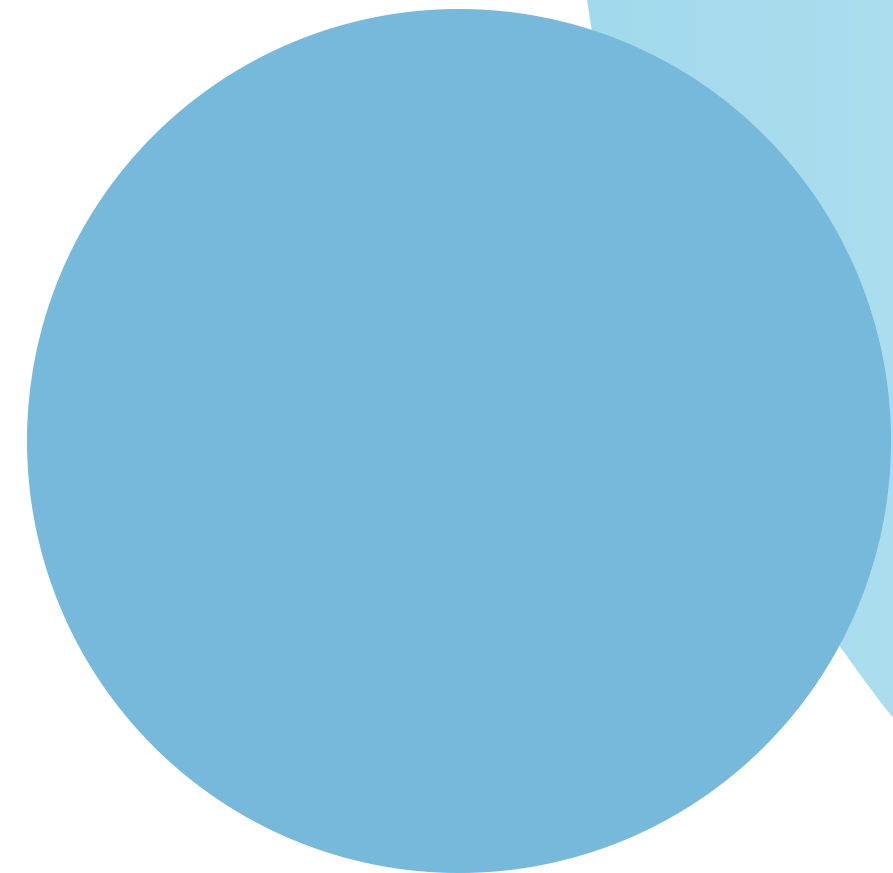
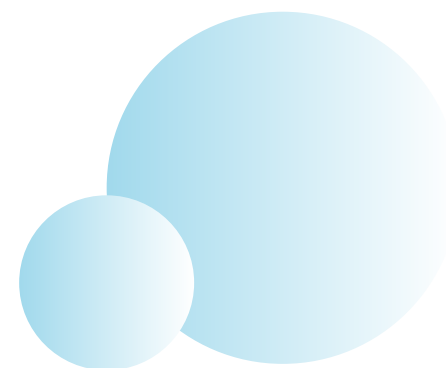
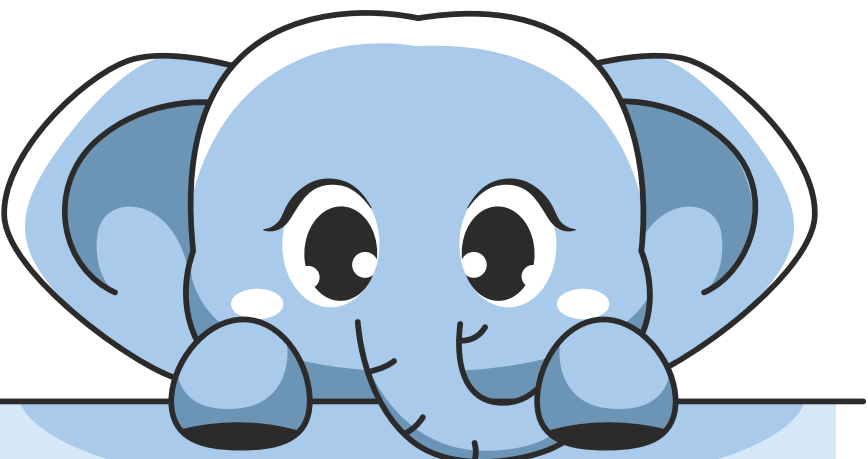
HOW TO USE ?

1. *Use cdn link*

2. `<script`

```
src="https://cdn.jsdelivr.net/npm/lodash@4.17.21/lodash.  
min.js"></script>
```

3. `const clone = _.cloneDeep(original);`



ASSIGNMENT

QUESTION FOR YOU

01 `const car = {
 brand: "Tesla",
 model: "X",
 features: {
 autopilot: true,
 color: "Red"
 }
};`

Task:

- Create both a shallow copy and a deep copy of car.
- Change the color in the shallow copy and deep copy.
- Show the difference in how they affect the original object.

Hint: For deep copy, use `structuredClone` or `JSON.parse(JSON.stringify(...))`



FOR ONLINE/OFFLINE SOFTWARE TRAINING



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